

DIRASCHOOL

Business Brief

School Management Information System



Purpose-built for the Kenyan Competency-Based Curriculum (CBC).

A production-ready, multi-tenant SaaS platform serving primary, junior secondary, and secondary schools across Kenya.

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For pricing strategy, investor conversations, and partnership discussions.

1 Executive Summary

DiraSchool is a cloud-based School Management Information System (SMIS) purpose-built for Kenyan schools operating under the Competency-Based Curriculum (CBC). It digitises and centralises every operational and academic function a school runs — from student admission through CBC report card generation — delivered as a multi-tenant Software-as-a-Service (SaaS) platform accessible from any browser, on any device, without installation.

The system is **production-ready, fully functional, and deployed on live cloud infrastructure**. It is not a prototype. Every module described in this brief exists and operates end-to-end today.

The business model is SaaS subscription, billed per school term — aligned to the Kenyan school calendar and school budget cycles, invoiced exactly when schools have money.

2 The Problem Being Solved

Kenyan schools — an estimated **35,000+ registered institutions** across primary, junior secondary (JSS), and secondary levels — manage their operations through:

- Physical registers and paper records
- Disconnected Microsoft Excel spreadsheets
- WhatsApp groups for staff communication
- Manual handwritten CBC report cards — a significant administrative burden given the 4-level rubric across multiple strands
- No parent visibility into fees owed, attendance, or results

The transition to CBC has made this worse, not better. CBC requires tracking competency levels (EE / ME / AE / BE) per subject per exam per student, generating narrative-style progress reports, and maintaining strand-level assessments — none of which Excel handles at scale. The consequences:

- Headteachers spending entire weekends on report card compilation
- Fee defaulters going undetected until term-end
- Attendance records lost or fabricated retrospectively
- No audit trail for administrative actions
- Parent complaints about lack of communication

DiraSchool eliminates all of these with one integrated platform.

3 Target Market

Primary Market

Private and semi-private primary and junior secondary schools in Kenya with 75–1,500 students, a functioning bursar/accountant, and teaching staff of 5–40. Urban and peri-urban schools in counties with reliable internet: **Nairobi, Mombasa, Kisumu, Nakuru, Eldoret, Thika, Nyeri, Meru.**

Secondary Market

- County government secondary schools seeking digital records compliance
- School chains and franchises (multi-campus operators) needing consolidated visibility
- Early Childhood Development Centres (ECDEs) as entry-level customers

Market Size (Conservative Estimate)

Segment	Estimated Count
Public primary schools	~23,000
Private primary schools	~8,000
Secondary schools (all types)	~4,000+
Addressable initial market (private primary + JSS, urban/peri-urban)	~4,000–6,000 schools

At an average of KES 22,000/term across the addressable market, the total addressable market (TAM) per term is **KES 88M–132M**.

4 What the System Does — Modules

4.1 Multi-Tenant School Onboarding

Each school on the platform is a fully isolated tenant. Data from School A is never visible to School B. A school is registered with its full legal details and assigned a subscription tier. A system superadmin manages all schools from a separate portal.

4.2 Role-Based Staff Management

Eight distinct user roles with carefully scoped permissions. Staff are invited by email with a secure one-time link. Admins can reset passwords, suspend or reactivate accounts without developer involvement.

4.3 Student Records Management

Complete student lifecycle: admission with full demographic capture, class assignment, status tracking (Active / Transferred / Withdrawn / Graduated), transfer history, bulk CSV import for migrating existing records, and full edit capability with audit trail.

4.4 Class Management

Class creation with name, stream, level category (Lower Primary / Upper Primary / JSS), and academic year. Class teacher and subject assignment. Year-end class promotion bulk-moves all students to the next class.

4.5 Attendance Management

Daily registers per class, auto-populated with all active students. Status per student: Present / Absent / Late / Excused. Draft-to-submitted workflow locks registers against retrospective changes. Data feeds directly into report card generation.

4.6 Subject & Examination Management

Subject creation, class assignment, and teacher assignment with HOD designation. Exam creation (Opener / Midterm / Endterm / SBA) with direct navigation to results entry.

4.7 Results Entry & CBC Grading

Bulk results entry per exam. Live CBC grade calculation as marks are typed — EE/ME/AE/BE for primary; EE1 through BE2 (8-point scale) for JSS. Pre-population of previously saved scores. Class statistics live as you type.

4.8 CBC Report Card Generation

Single student or entire class generation in one action. System automatically pulls results, computes weighted-average percentages per subject, derives CBC grades, aggregates overall grade, and summarises attendance. Per-subject and overall remarks editable before publication. Publication locks the card permanently. Print template: A4-formatted page with school header, student details, subject table, attendance summary, CBC grading key, remarks, and three-party signature lines — auto-triggers browser print/PDF dialog.

4.9 Fee Management

Named fee structures per academic year and term with line items and auto-computed totals. Payment recording with method tracking (cash, M-Pesa, bank, cheque). Running balance per student. Payment reversal with reason. Accountant-only access to financial operations.

4.10 Timetable, Library & Transport

Weekly timetable grids per class with teacher and room assignment. Book catalogue, loan issuance, return recording, and overdue tracking. Transport route management with student assignment.

4.11 Audit Logging

Every significant action logged: who, what, which resource, when, and what changed. School admins review their own logs. Superadmin views logs across all schools — critical for compliance, dispute resolution, and fraud detection.

4.12 Parent Portal

Read-only portal scoped to the parent's own children: profile, fee balance and payment history, attendance, exam results, and published report cards.

4.13 Superadmin Portal

Separate management interface for DiraSchool operators: school listing with student and staff counts, subscription management (plan tier, status, trial expiry), and system-wide audit log viewer with filters.

Staff Role Matrix

Role	Access Level
School Admin	Full access to all modules
Director	Full access
Head Teacher	Full access including audit logs
Deputy Head Teacher	Academic and operational access
Teacher	Students, attendance, results, timetable
Secretary	Students, attendance, classes, fee view
Accountant	Fee structures, payments, financial reports
Parent	Read-only portal: own child's data only

5 How the System Is Built

5.1 Technical Architecture

DiraSchool is a decoupled full-stack SaaS application with a REST API backend and a React frontend, deployed independently on DigitalOcean cloud infrastructure. Both servers are managed with **Nginx** as the reverse proxy and **PM2** as the Node.js process manager — a production-grade setup providing SSL termination, process auto-restart, zero-downtime reloads, and cluster-mode CPU utilisation.

5.2 Backend Stack

Component	Technology	Purpose
Runtime	Node.js 20 LTS	Server runtime
Framework	Express.js	HTTP routing and middleware
Database	MongoDB (Mongoose ODM)	Primary data store — 14 collections
Authentication	JWT (HTTP-only cookies)	Session management
Email Queue	BullMQ + Redis	Async email delivery
Email Service	ZeptoMail (Zoho)	Transactional email — invites, resets, notifications
Validation	Zod	Request schema validation — rejects malformed input
Password Hashing	bcrypt (cost factor 12)	Credential security
File Handling	Multer	CSV import for bulk student upload
Audit Logging	Custom middleware	Action tracking across all modules

5.3 Frontend Stack

Component	Technology	Purpose
Framework	Next.js 15 (App Router)	React framework, file-based routing
Language	JavaScript (JSX)	Component development
Styling	Tailwind CSS	Utility-first CSS
Component Library	shadcn/ui	Accessible, composable UI components
Data Fetching	TanStack Query v5	Server state, caching, background refetch
Forms	React Hook Form + Zod	Client-side validation
Auth State	Zustand	Lightweight client state store
Notifications	Sonner	Toast notification system

5.4 Infrastructure (DigitalOcean Droplets)

Both servers run on DigitalOcean Droplets with **Nginx** as the reverse proxy and **PM2** as the process manager. Nginx handles SSL/TLS termination via Let's Encrypt (Certbot), reverse-proxies requests to Node.js, serves static assets directly, enforces rate limiting, and manages HTTP-to-HTTPS redirects. PM2 manages the Node.js process lifecycle — auto-restart on crash, cluster mode across CPU cores, log rotation, and zero-downtime deploys via *pm2 reload*.

Component	Service	Specification	Monthly (USD)
API Server	DO Droplet	2GB RAM / 1vCPU / 50GB SSD	\$12

Web Server	DO Droplet	1GB RAM / 1vCPU / 25GB SSD	\$6
Database	MongoDB Atlas Flex	Scales with usage, managed	\$0–\$58
Object Storage	DO Spaces	S3-compatible, 250GB included	\$5
Backups	DO Snapshots	Weekly automated droplet snapshots	\$2–\$4
Email	ZeptoMail	Transactional, pay-per-use credits	~\$2–\$5
SSL / HTTPS	Let's Encrypt (Certbot)	Managed via Nginx, auto-renews	Free
Total (early stage)			~\$27–\$40/month
Total (50+ schools)			~\$100–\$180/month

5.5 Email Service: ZeptoMail

ZeptoMail (by Zoho) is the recommended transactional email provider. It is purpose-built exclusively for transactional email — not shared with marketing campaigns — keeping IP reputation high and ensuring inbox delivery for critical messages: staff invites, password resets, and parent notifications.

Attribute	Detail
Provider	ZeptoMail by Zoho
Use cases	Staff invites, password resets, email verification, parent notifications
Free tier	10,000 emails (one-time, sufficient for early onboarding)
Paid pricing	~\$2.50 per 10,000 email credits
Integration	REST API and SMTP relay — Node.js via Nodemailer transport
Deliverability	High — dedicated transactional infrastructure
Alternative	Resend — modern API, 3,000 free emails/month, excellent DX
Est. annual cost (Year 1)	Below KES 500/month

5.6 Security Architecture

Layer	Implementation
Session tokens	HTTP-only cookies — not accessible to JavaScript; eliminates XSS token theft
Passwords	bcrypt hashed at cost factor 12
Route protection	Role-based middleware on every protected endpoint
Tenant isolation	schoolId scoped on every DB query — tenants cannot access each other's data
Input validation	Zod schema validation on all API inputs — rejects malformed/unexpected data
Password reset	Time-limited secure tokens, single-use
Audit trail	Every significant action logged with actor, timestamp, and change metadata
Transport	HTTPS enforced via Nginx; HTTP redirects to HTTPS automatically

6 Competitive Landscape in Kenya

Competitor	Strengths	Weakness vs. DiraSchool
Zeraki	4,000+ schools, brand recognition, analytics	Heavy, expensive, not CBC-native from ground up
Schoolap	Parent communication focus	Weak academic and fee management
Shule Pro	Basic attendance and fees	No CBC report cards, limited modules
Elimu	Government-adjacent	Limited private school features
Excel + paper	Free, familiar	The status quo being replaced — not a real competitor

DiraSchool's Key Differentiators

- **CBC-native from day one** — grading logic, report cards, and terminology built around CBC, not retrofitted onto an older system.
- **Per-term billing** — aligned to when schools actually have money, not a monthly invoice during the school holidays.
- **One platform, all modules** — no integration headaches, one login, one support contact for the whole school.
- **Parent portal included at no extra charge** — most competitors charge extra for parent access.
- **Transparent audit trail** — every action logged; headteachers see exactly who did what and when.
- **Built entirely for Kenya** — KES currency, Kenyan school calendar, county fields, CBC rubric, local school terminology throughout.

7 Pricing Structure

All prices in Kenyan Shillings. **VAT at 16% applicable on all tiers.** Per-term billing is the primary model — aligned to school fee collection cycles. Monthly billing is available at a 20% premium. Annual billing offers approximately 15% off (one term effectively free).

Per-Term Pricing (Primary Model)

Plan	Students	Per Term (ex-VAT)	Per Term (incl. VAT)	Annual (ex-VAT)	Annual Saving
Dira Seed	Up to 75	KES 6,500	KES 7,540	KES 16,500	~KES 3,000
Dira Lite	Up to 200	KES 12,500	KES 14,500	KES 32,000	~KES 5,500
Starter	Up to 400	KES 20,000	KES 23,200	KES 51,000	~KES 9,000
Growth	Up to 850	KES 45,000	KES 52,200	KES 115,000	~KES 20,000
Professional	Up to 1,500	KES 75,000	KES 87,000	KES 190,000	~KES 35,000
Enterprise	Unlimited	Custom	Custom	Custom	—

Trial: 30 days free, up to 50 students, no credit card required. Schools upgrade to a paid plan when they are ready — no automatic billing.

Cost Per Student Per Term

Plan	Students	Cost per student / term	Comparable cost
Dira Seed	75	KES 87	2 exercise book pages
Dira Lite	200	KES 63	Less than a biro pen

Starter	400	KES 50	Less than a biro pen
Growth	850	KES 53	Less than a biro pen
Professional	1,500	KES 50	Less than a biro pen

DiraSchool as % of School Fee Income

Plan	Students	Est. Term Fee Income	DiraSchool Cost	As % of Income
Dira Seed	75	KES 375K–1.1M	KES 6,500	0.6–1.7%
Dira Lite	200	KES 1M–3M	KES 12,500	0.4–1.3%
Starter	400	KES 2M–6M	KES 20,000	0.3–1.0%
Growth	850	KES 4.25M–12.75M	KES 45,000	0.4–1.1%
Professional	1,500	KES 7.5M–22.5M	KES 75,000	0.3–1.0%

* Fee income estimate: KES 5,000–15,000 per student per term depending on school type and location.

8 Business Model Summary

Metric	Detail
Revenue model	B2B SaaS, term-based subscription
Primary billing cycle	Per Kenyan school term (3 times per year)
Secondary billing cycles	Annual (15% discount) / Monthly (20% premium)
Customer acquisition	Direct outreach, head-teacher networks, KNUT/KUPPET, school expos
Onboarding	Self-service 30-day trial → sales-assisted conversion
Support model	WhatsApp + email support; in-app documentation (planned)
Churn mitigation	Term-aligned billing, sticky historical data, high switching cost
Expansion revenue	Automatic plan upgrades as school enrolment grows past tier limit
Break-even point	5–8 paying schools at Starter tier average
Target Year 1	30 paying schools
Target Year 2	100 paying schools

Revenue Projections

Schools	Plan Mix	Revenue / Term	Infra / Term	Net / Term	Net / Year
10	5 Lite + 3 Starter + 2 Growth	KES 218,500	KES 20,000	KES 198,500	KES 595,500
20	Mixed across tiers	KES 490,000	KES 30,000	KES 460,000	KES 1.38M
40	Mixed across tiers	KES 980,000	KES 55,000	KES 925,000	KES 2.775M
80	Mixed across tiers	KES 1,960,000	KES 100,000	KES 1,860,000	KES 5.58M

Invoice Timing Strategy

Invoices sent two weeks before each term starts, payment due by the first day of term — when schools have just processed parent fee collections.

Term	Schools Open	Invoice Send Date	Payment Due
Term 1	~2nd week January	Late December	January 6
Term 2	~1st week May	Mid-April	April 28
Term 3	~1st week September	Mid-August	September 1

9 The One-Paragraph Pitch

DiraSchool is the only school management system in Kenya designed from the ground up for CBC — not adapted from an older curriculum system, not a generic African school platform retrofitted with a CBC label. It handles everything a school runs: student records, daily attendance, CBC report cards with the correct 4-level and 8-point grading rubrics, fee collection, exam results, staff management, a parent portal, and a full audit trail — in one login, on any device, at a price that represents less than 1% of what the school earns in fees each term. It bills per school term because that is when schools have money, not monthly when they are in the middle of the holidays with no income.